# 05 NEURO-TREAD

Bio-inspired raised neural network over a dimpled adaptive base.

**66.1 g**

PLA · 100 % infill · sealed core

✓ LIGHTENED CORE · SLOT + HOLE



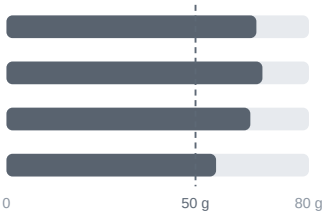
3/4 view · rendered from the STL, colours are preview only

## AT A GLANCE

| | |
|---|---|
| Overall height | **135.00 mm** |
| Max footprint (foot) | **42.63 × 36.61 mm** |
| Paddle slot | **30.2 × 12.2 mm, 130.0 deep** |
| Textured band | **z 21.5 – 116.0 mm (94.5 tall)** |
| Outer size in band (max) | **39.1 × 33.0 mm** |
| Peak texture height | **3.50 mm above base** |
| Screw hole (butt) | **Ø 3.5 mm + Ø 6.2 × 2.2 counterbore** |
| Material volume / surface | **53.30 cm³ / 458 cm²** |

## WEIGHT · 100 % INFILL

| | | |
|---|---|---|
| **PLA** | 1.24 g/cm³ | **66.1 g** |
| **PETG** | 1.27 g/cm³ | **67.7 g** |
| **TPU** | 1.21 g/cm³ | **64.5 g** |
| **ABS** | 1.04 g/cm³ | **55.4 g** |




## COLOURWAY



#18181A #CE1820

3MF print palette

## PRINT PROFILE

| | |
|---|---|
| Infill / layer height | **100 % / 0.20 mm** |
| Layers | **675** |
| Orientation | **upright, butt end down** |
| Supports | **none required** |
| Sealed hollow core | **33.6 cm³, walls ≈ 1.2 mm** |
| Filament, 1.75 mm | **≈ 22.2 m (100 % flow)** |

## DESIGN INTENT

A network of 48 neurons joined by curved axon ridges rises to the full 3.5 mm crest, standing proud of a field of small bumps. The ridges guide the fingers while the bump field provides fine grip.

*Ridges and bumps are gentle slopes; the network is printed in the same material as the base.*

## KEY FEATURES

- 48 somas (8 × 6) joined by curved axons about 1.5 mm wide
- Bump field on a 2.66 mm hex pitch
- Network and somas reach the full 3.5 mm crest

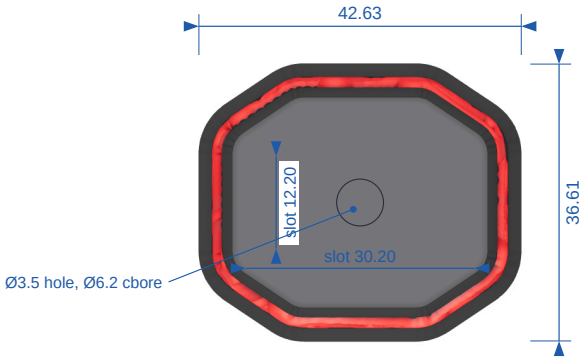
## VERIFIED ON THE STL

| | |
|---|---|
| Paddle slot | **30.2 × 12.2 mm (30 × 12 + 0.2)** |
| Screw hole + counterbore | **Ø 3.5 mm + Ø 6.2 × 2.2** |
| Outer envelope vs reference | **+0.000 mm at most** |
| Outer body / sealed cavity | **1 / 7** |
| Wall around the cavity (min) | **1.10 mm** |
| Material under 0.4 mm wide | **0.004 % of volume** |
| Widest flat bridge | **3.6 mm** |

## FILES

| | |
|---|---|
| stl/05_neuro_tread.stl | 24.5 MB · 489,688 triangles |
| 3mf/05_neuro_tread.3mf | 5.2 MB · colour |

✓ SLOT & HOLE VERIFIED ✓ ONE BODY · SEALED CORE ✓ PRINT-READY · 0.20 MM LAYERS ✓ 3MF VALIDATED

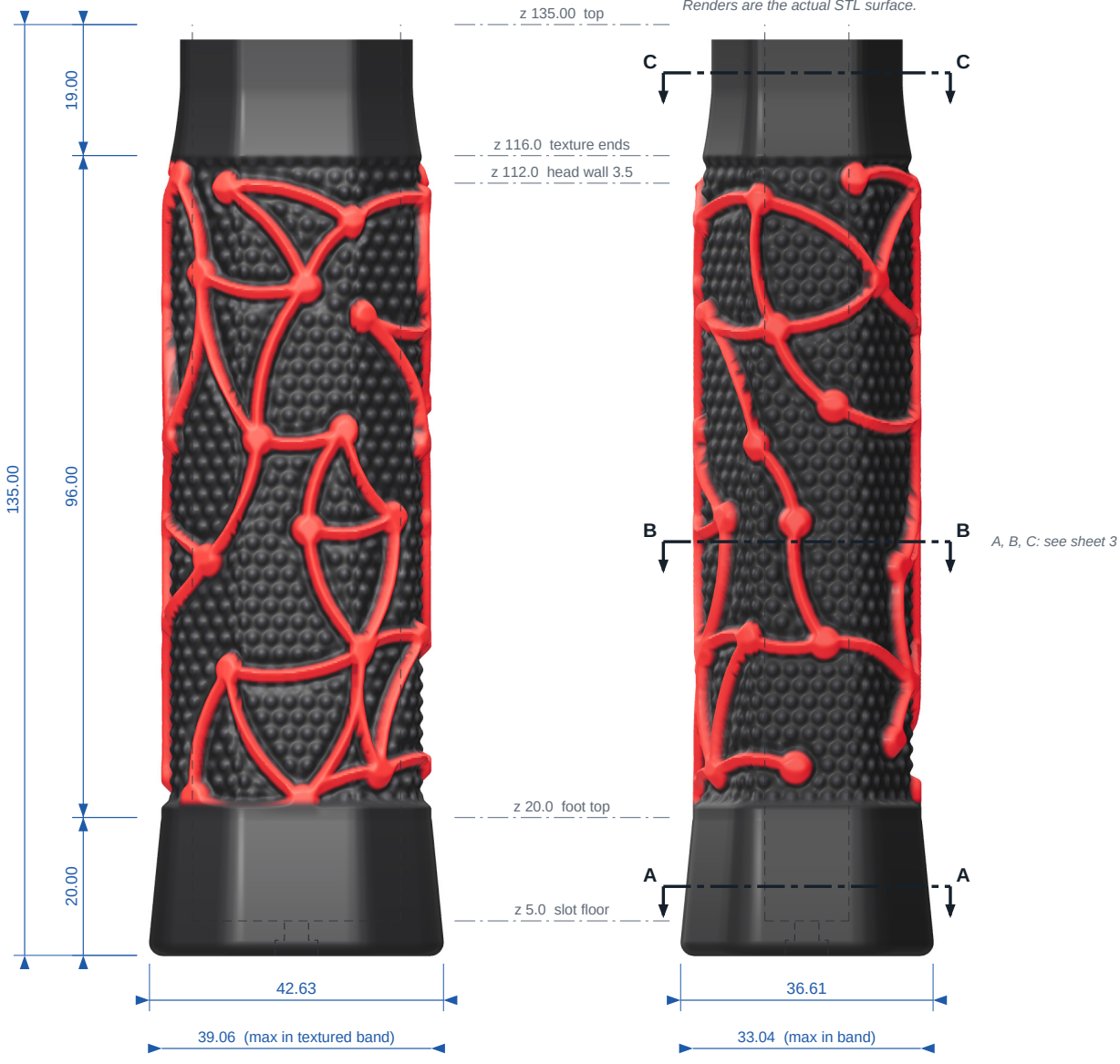



TOP VIEW (from +Z)

## KEY DIMENSIONS

| | |
|---|---|
| Overall height | **135.00** |
| Overall width × depth (foot) | **42.63 × 36.61** |
| Paddle slot width × depth | **30.20 × 12.20** |
| Slot depth (floor at z 5.0) | **130.00** |
| Outer W × D, textured band (max) | **39.06 × 33.04** |
| Foot top (z 20) | **39.16 × 33.13** |
| Butt face (z 0) | **40.56 × 34.54** |
| Peak texture height above base | **3.50** |
| Screw hole / counterbore | **Ø 3.50 / Ø 6.20 × 2.2** |

*All values in mm. Dashed lines: paddle slot and screw hole (hidden edges). Renders are the actual STL surface.*




FRONT VIEW (from −Y)

RIGHT VIEW (from +X)

**SCALE 1 : 1**




PART

# NEURO-TREAD

Pickleball paddle grip sleeve, closed butt, open throat

GENERAL NOTES

1. Views are true size: print this sheet at 100 %, do not fit to page.
2. Dimensions are measured from the STL, not nominal.
3. Sealed hollow core, walls ≈ 1.2 mm; head above z 100 solid, end walls slit 20 mm.
4. Outer envelope matches PickleballGrip_1.3mf below z 90; head thickened.

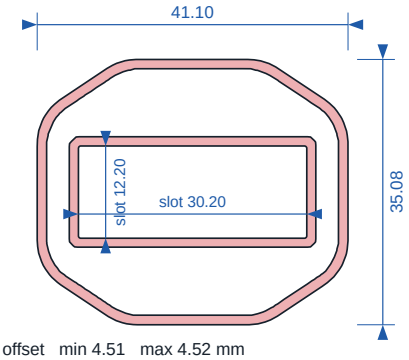
| | | |
|---|---|---|
| SCALE **1 : 1** | UNITS **millimetres** | PROJECTION |
| MATERIAL **PLA / PETG / TPU** | INFILL **100 %** | LAYER HEIGHT **0.20 mm** |
| MASS (PLA) **66.1 g** | VOLUME **53.30 cm³** | SURFACE AREA **458 cm²** |
| REVISION **7** | DATE **2026-10-22** | SHEET **2 of 4** |

# Sections & profiles

True-size cross-sections, how the outer surface and material change with height, and the dimension tables.

### SECTION A–A

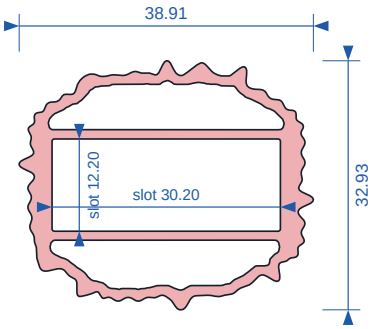
z = 10 mm · foot




Skin offset min 4.51 max 4.52 mm
Area 268 mm² · outer perimeter 128.9 mm

### SECTION B–B

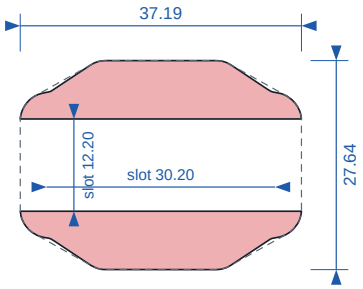
z = 60 mm · textured band




Skin offset min 1.41 max 3.46 mm
Area 293 mm² · outer perimeter 116.9 mm

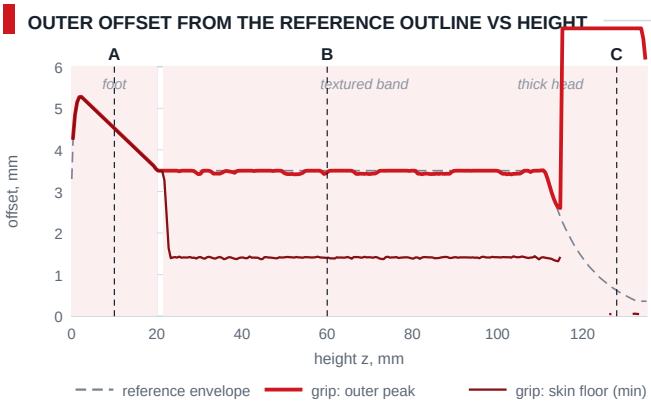
### SECTION C–C

z = 128 mm · rim

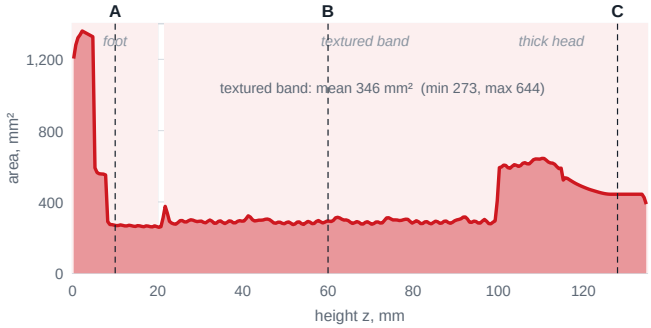



Skin offset max 6.92 mm
Area 443 mm² · outer perimeter 110.7 mm

## OUTER OFFSET FROM THE REFERENCE OUTLINE VS HEIGHT




## MATERIAL CROSS-SECTION AREA VS HEIGHT




## OVERALL & SLOT

| | |
|---|---|
| Overall length (Z) | **135.00 mm** |
| Overall width (X) | **42.63 mm** |
| Overall depth (Y) | **36.61 mm** |
| Slot width × depth | **30.20 × 12.20 mm** |
| Slot area / perimeter | **368.4 mm² / 84.3 mm** |
| Slot depth (floor at z 5.0) | **130.00 mm** |
| Screw hole / counterbore (butt) | **Ø 3.50 / Ø 6.20 × 2.2** |
| Sealed cavity / min wall | **33.6 cm³ / 1.10 mm** |
| Outer W × D in band (max) | **39.06 × 33.04 mm** |
| Material volume / surface | **53.30 cm³ / 458.4 cm²** |

## ZONES BY HEIGHT

| ZONE | Z FROM | Z TO | HEIGHT |
|---|---|---|---|
| Solid floor (butt) | 0.0 | 5.0 | **5.0** |
| Flared foot | 0.0 | 20.0 | **20.0** |
| Smooth collar | 20.0 | 21.5 | **1.5** |
| Textured band | 21.5 | 116.0 | **94.5** |
| Smooth neck | 116.0 | 112.0 | **-4.0** |
| Head (solid, thick wall, slot opening) | 112.0 | 135.00 | **23.0** |

| | |
|---|---|
| Relief floor above reference line (design / min) | **1.20 / 1.37 mm** |
| Slot wall over the head (design / min) | **3.50 / 3.52 mm** |
| Peak texture height above base | **3.50 mm** |

## OUTER DIMENSIONS BY HEIGHT

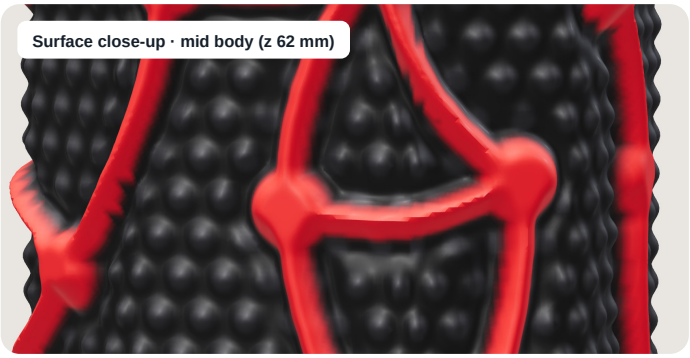
| Z (MM) | OUTER W | OUTER D | PERIMETER | AREA MM² | SKIN MIN | PEAK OFFSET |
|---|---|---|---|---|---|---|
| **5.0** | 42.08 | 36.06 | 132.0 | 959 | 5.00 | 5.01 |
| **10.0** | 41.10 | 35.08 | 128.9 | 269 | 4.51 | 4.52 |
| **20.0** | 39.11 | 33.09 | 122.6 | 260 | 3.52 | 3.52 |
| **30.0** | 37.77 | 31.75 | 116.8 | 292 | 1.42 | 3.42 |
| **45.0** | 38.99 | 32.88 | 117.1 | 301 | 1.40 | 3.50 |
| **60.0** | 38.91 | 32.93 | 116.9 | 293 | 1.41 | 3.46 |
| **75.0** | 38.94 | 32.96 | 119.3 | 302 | 1.41 | 3.50 |
| **90.0** | 38.99 | 32.86 | 117.2 | 297 | 1.42 | 3.50 |
| **105.0** | 38.81 | 32.79 | 116.8 | 602 | 1.42 | 3.43 |
| **116.0** | 37.20 | 30.19 | 114.2 | 534 | open | 6.92 |
| **125.0** | 37.19 | 27.75 | 110.8 | 447 | open | 6.92 |
| **132.0** | 37.19 | 27.64 | 110.7 | 443 | 0.06 | 6.92 |

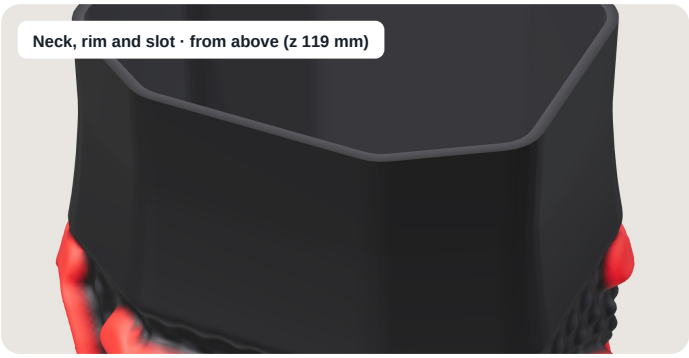
# Views & surface

Four-way turntable, close-ups of the surface, and the unrolled height map with the design parameters.

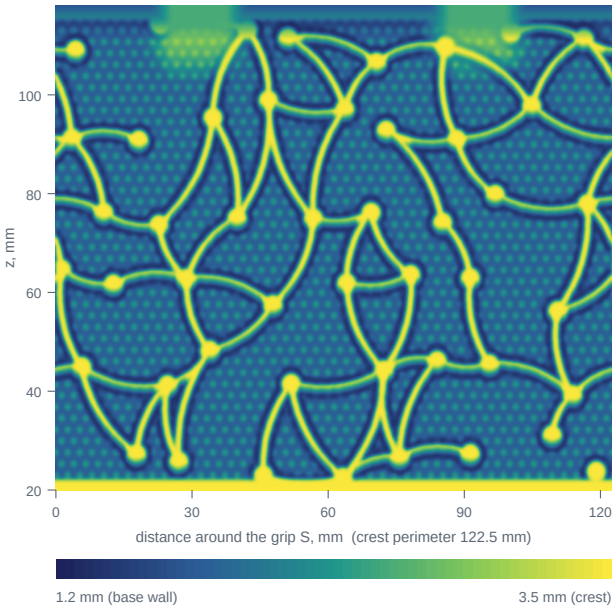


Four views, 90° apart · rendered from the STL, colours are preview only




## UNROLLED SURFACE · HEIGHT ABOVE BASE




## DESIGN PARAMETERS

| | |
|---|---|
| Neurons (somas) | **48 (8 × 6)** |
| Axon ridge width | **about 1.5 mm** |
| Soma diameter | **about 4.2 mm** |
| Network height | **3.50 mm above base** |
| Bump pitch | **2.66 mm (hex)** |
| Bump field height | **1.4 – 2.4 mm above base** |
| Ridge / bump colours | **red / black** |

## PRINT & VERIFICATION

| | |
|---|---|
| Result: slot & hole / print-ready | **PASS / PASS** |
| Outer body · genus · sealed cavity | **1 · 1 · 7** |
| Wall around the cavity (min) | **1.10 mm** |
| Paddle slot | **30.2 × 12.2 mm** |
| Envelope over reference | **+0.000 mm** |
| Material under 0.4 mm wide | **0.004 %** |
| Widest flat bridge | **3.6 mm** |